Project 2
CIT1C21 - Application Development Project (65%)
CIA1C06 - Database Application Development (55%)
(System Code Implementation)

Introduction

This assignment aims to assess students' understanding and practical application of web development principles. The objective is to design and implement a functional website by creating both front-end and back-end components.

Learning outcomes:
By the end of this assignment, you should be able to:

- Implement the frontend of a website using web technologies (HTML, CSS, JavaScript).
- Implement the backend of a website, including server-side logic and database integration.

Notes:

a. This is an individual assignment.
b. Your solution should cover the requirements above.

Using solutions found online or AI generated work from the internet is prohibited.

Report Template:
A **report template**, labeled 'ADEV DBAV-AY2526 Project 2 Template', will be provided for you to complete. In the template, you will need to:

ADEV

- List down all the URLs to the respective pages and provide sample input.
- Provide screenshot(s) of the user interfaces of the main features.
- Provide screenshot(s) of your additional feature and explain how it enhances the website.

DBAV

- List all the SQL statements, route pattern, HTTP method and payload (if any) for the REST APIs implemented.
- Explain the error-handling mechanisms that were implemented.

Note: Your tutor will only grade the web pages and API routes listed in the report. Therefore, it is your responsibility to ensure that all the required information is included.

Project Brief

Animal Adoption Portal – Temasek Polytechnic, Diploma in Veterinary Technology

This website provides a comprehensive listing of all animals available for adoption through the diploma's adoption initiative. The portal serves as both an information hub for the public to browse animals up for adoption, and as a management platform for administrators to manage adoption listings.

Your role is to provide the admin of the portal with the functionality to view, create, update, and delete adoptable animal listings.
The website should allow the following operations:

- **Display all adoptable animals**

Create a main page that displays all animals available for adoption from the database.
The following information should be displayed for each animal:

- The species (e.g., dog, cat, rabbit - You should have at least a minimum of 3 different species)
- Breed (e.g., German Shepherd, Siamese)
- Age (e.g., 1 year and 2 months)
- Adoption status (Available, Reserved, Adopted)
- A hyperlink or modal that displays more detailed information about the animal

- **Display an animal detailed profile**

When the user clicks on the animal link on the main page, open a new page or modal showing a detailed animal profile.
The following information should be displayed:

- Animal name
- Image(s) of the animal
- Age
- Species
- Breed
- Gender
- Personality/Temperament description

- Adoption status

- **Add, update and delete animal listings**

The admin should have privileged access to manage the database of adoptable animals.
When adding a new animal listing, the following information must be input:

- Animal name
- Image(s) of the animal
- Age
- Species
- Breed
- Gender
- Personality/Temperament description
- Adoption status (Available, Reserved, Adopted)

The admin should also be able to update or delete existing listings as needed.

ADEV Part 1: System Code Implementation (45%)

Frontend Development

You are tasked with developing the website's frontend capabilities. The website **must make use of the RESTful APIs implemented in the DBAV section** of this project. Your design must incorporate essential elements and demonstrate a clear connection between your frontend application and database.

You have the freedom to design the web pages in the style of your choice, but you will need to ensure the project requirements are met.

- ### Frontend Features Implementation (25%)

Implement the core features to allow users to access and utilize the website's features to perform the required tasks. Your website must cover the following features:

- Display all adoptable animals
- Display an animal detailed profile
- Add an animal listing
- Update an animal listing
- Delete an animal listing

Additionally, your implemented features are expected to provide consistent and reliable error handling to maintain smooth user interactions.

You are required to list all the URLs to the different pages of the website in the report and provide sample input values if applicable.

- ### User Interface (10%)

Implement a user interface that is cohesive, intuitive, easy to navigate, and appropriate for the intended user.

- **Appropriate and cohesive design:** Use the same colors, fonts, and layouts across all pages which are suitable for the project theme.
- **Easy to navigate:** Menus, buttons, and links should be clear and in logical places.

- **User-friendly:** Core features like viewing, adding, or editing animals should be simple and intuitive.

Provide screenshot(s) of the implemented user interface for each of the core features (list animals, view details, add, update, delete) in your report.

- ## Additional Feature (5%)

You are encouraged to enhance the website with an extra feature beyond the basic requirements. The addition should improve the **functionality** of the site. **Only one** additional feature will be considered for grading.

The feature should show thoughtful effort (e.g., simply changing colors or adding text is not sufficient). Examples of eligible additional feature:

- **Functionality Add-ons**: Contact form, search functionality, basic authentication

You are to implement an additional feature which can fulfill both ADEV and DBAV requirements for Project 2. The implemented feature must be presented during your demonstration and documented in the report. You have a 150-word limit for your explanation.

- **Process (5%)**

You will be assessed on your **progress** throughout the project and how you **communicate effectively** with your tutor. You are expected to:

- Provide **clear evidence of progress**, showing steady work toward your goals rather than last-minute effort.
- Communicate **proactively and with clarity** to your tutor in class on your progress (What did you do? What are you working on? What obstacles are you facing? What is your next step?) in class during Week 14 (Week of 19 Jan) and Week 16 (Week of 2 Feb).

Note: you do not need to include this process section in your report.

DBAV Part 2: System Code Implementation (35%)

You are tasked with coding the backend of your web application to provide RESTful API for your frontend code. You are free to design your database and to populate the database with meaningful data. Your database design must contain at least two related tables with foreign keys (For eg. Pet and Image entities)

- **API Implementation (20%)**

Demonstrate API implementations of the SQL queries for all required operations. There should be at least one SQL query for Create, Update, Retrieve, and Delete operations.

Include all SQL statements, route patterns, HTTP methods, and payloads (if any) in the report.

- **Additional Feature (5%)**

Based on the additional feature you have identified and designed, you are tasked to link the frontend implementation to the backend. Provide the RESTful API implementation.

Include all SQL statements, route patterns, HTTP methods, and payloads (if any) in the report.

- **Error Handling (10%)**

You are tasked to implement 2 (TWO) error-handling mechanisms on your server-side system, explaining how you carried out the implementation and why this is important. Provide examples to support your answer

You have a 200-word limit per mechanism.

ADEV & DBAV Part 2: Demonstration (ADEV – 20%, DBAV – 20%)

Besides submitting your **project source code** and **written report**, you are required to **present and demonstrate your work** during the assignment evaluation in **Week 17 (week of 9 Feb, timings to be advised by your tutor).**

During the evaluation, you are expected not only to show that your project runs correctly, but also to demonstrate a clear understanding of the implementation details, design choices, and rationale behind your solution in the interview section.

This means you should be able to:

- Walk through the structure of your code and explain how the different components work together.
- Justify the logic and approach you used to solve the problem (e.g., why a certain algorithm, function, or library was chosen).
- Respond confidently to technical questions from your tutor about your code and solution.

Presentation Requirement: You must present **within the allocated time limit** (to be advised by your tutor).

Submission

You are to submit the report, project, and the SQL scripts. Submit the deliverables into LMS under the Assessment folder by **Week 17, Monday, 9th Feb 2026, at 0800hrs (8 am).**

Submission instructions:

1. Make a copy of your project.
2. In the copy, remove the node_modules folder.
3. Export SQL scripts of your database and name it as project2.sql. (Make sure data is also exported)
4. Put your project, SQL script and the report into the same folder and zip up the folder.
5. You should name your folder according to the following format:

   StudentID-YourName-ADEV-DBAV-AY2526-Oct-Project 2.zip

e.g.: 1234567D-Teo_Chu_Chu-ADEV-DBAV-AY2526-Oct-Project 2.zip

When submitting assignments, each student is responsible for ensuring that they do not submit an empty zip file or a corrupted file.
Students must also ensure they are using a suitable device when working on assignments / projects, maintain proper backups of their work and do not cite 'laptop issues' as a reason for assignment extensions as such reasons will not be considered.

Penalty for Late Submission

late and < 1 day : 10% deduction from absolute mark given for the assignment
late >= 1 and < 2 days : 20% deduction from absolute mark
late >= 2 days : No marks awarded

Note that "day" includes **non-working days** (Sat, Sun and public holidays).

Project – Grading Criteria

The grading criteria for **ADEV Part 2: System Code Implementation (65%)** will be based on the following:

Criteria	In Context	Performance Level				
		A (>=80%)	B (>=70%)	C (>=60%)	D (>=50%)	F (<50%)
		Excellent	Good	Average	Pass	Below Standard
Process (5%)	Progress & Communication (5%)	Clear evidence of substantial progress; milestones well-defined with appropriate timeline; communicates proactively and with clarity about completed work and next steps.	Evidence of steady progress; milestones defined; communicates effectively about completed and upcoming work.	Some progress shown; milestones vague or incomplete; communicates when prompted but not fully clear or proactive.	Minimal progress evident; milestones unclear or missing; communication limited or unclear.	No measurable progress OR absent from check-in without valid reason.
Frontend Implementation (25%)	Completeness of features (20%)	All the required website features are fully implemented and functional. Demonstrates strong understanding of code implemented during the demonstration by clearly explaining key logic, purpose, and/or behaviour.	All the required website features are fully implemented and functional. Demonstrates some understanding of code implemented during the demonstration.	Only a portion of the website features are implemented and functional. Demonstrates limited understanding of code implemented during the demonstration.	A minimal number of the required website features are implemented and functional. Struggles to demonstrate understanding of code implemented during the demonstration.	Non-submission; or no features are functional; or no evidence of understanding of code implemented during the demonstration

	Robustness of frontend (5%) (Robustness means to prevent error and remain functional in the event of error)	The website consistently handles requests under typical conditions and recovers well from common errors or abnormal inputs. Mechanisms to ensure valid operations are highly reliable and consistently effective, preventing almost all errors.	The website handles standard operations without crashing but may not recover from common errors. Mechanisms to ensure valid operations are effective in most scenarios, minimizing errors during normal use.	The website handles standard operations without crashing but may not recover from common errors. Mechanisms to ensure valid operations are present but inconsistently applied, leading to occasional errors.	The website frequently crashes or returns errors during normal operations. Mechanisms to ensure valid operations are weak, leading to common errors or unexpected results.	Non-submission or no measures are considered to make the website robust.
User Interface (10%)	Intuitive user interface (5%) - Ease of use	The interaction flow is seamless and user-friendly, allowing users to move effortlessly between sections and complete tasks without confusion.	The interaction flow is generally smooth, though some areas could benefit from clearer organization or labels. Most actions are easy to figure out	The interaction flow works but may require extra effort from users. Some elements are unclear or difficult to locate, reducing the ease of use.	The interaction flow is confusing and disorganized, making it difficult for users to find features or complete tasks efficiently.	The interaction flow is broken and highly confusing with non-functional elements that require constant page transitions and prevent task completion.
	Visual design (5%) - aesthetics	The design is polished and professional, with a	The design is appealing, with a well-organized	The design feels basic and inconsistent, with elements that lack	The design lacks quality, with poorly arranged or	The design appears

	- coherence	cohesive and harmonious overall presentation. Elements are arranged thoughtfully to create a visually engaging and readable interface.	structure, though there are minor inconsistencies in presentation.	harmony and attention to detail, leading to a less engaging appearance.	mismatched elements that make the interface confusing or unattractive.	incomplete or unprofessional. Crucial elements missing, making the website unusable.
Additional Feature (5%)	Implementation (5%) - value	The additional feature implemented is bug-free, fully functional, and provides exceptional value to the application, greatly enhancing its overall functionality or user experience.	The additional feature is bug-free and provides significant value to the application, enhancing its overall functionality or user experience.	The additional feature has no major bug and provides moderate value to the application, partially enhancing its overall functionality or user experience.	The additional feature is buggy and/or provides minimal value to the application and do not enhance its overall functionality or user experience.	Non-submission or failure to add any meaningful functionality or user experience improvement.
Demonstration & Code Interview	Q&A (20%)	Answers all technical/code-related questions confidently; gives a coherent explanation of logic and rationale behind technical solution.	Answers most technical/code-related questions confidently; gives reasonable explanation of logic and rationale behind technical solution.	Struggles with some technical/code-related questions; explanations of technical solution lacks depth or clarity.	Unable to respond to most technical/code-related questions; shows limited understanding of own technical solution.	Student was absent for the project demonstration without a valid reason. Clear evidence of plagiarism

		Present within the given time-limit.	Present within the given time-limit.			detected (disciplinable offence)

The grading criteria for **DBAV Part 2: System Code Implementation (55%)** will be based on the following:

Criteria	In Context	Performance Level				
		A (>=80%)	**B (>=70%)**	**C (>=60%)**	**D (>=50%)**	**F (<50%)**
		Excellent	**Good**	**Average**	**Pass**	**Below Standard**
API Implementation (20%)	Backend implementation (20%)	All the website's backend requirements are met, with the API consistently handling requests under typical conditions			Part of the website's backend requirements are met, with the API handling some requests under typical conditions	No database operation is present Incorrect SQL Codes, Implementation failed to function, Non-submission of Part 2
		Proper SQL implementation for CRUD operations, including the use of multiple JOIN queries	Proper SQL implementation of CRUD operations, including the use of at least one join query.	Correct SQL implementation of CRUD operations	Partially correct SQL implementation of CRUD operations	
		Populated data contains comprehensive set of	Populated data contains good set of meaningful records	Populated data contains good set of meaningful records	Populated data contains very few records	

		meaningful records				
Additional Feature (5%)	Implementation of Additional Features from ADEV Part 2 (5%)	The additional feature's backend API implementation is provided and fully functional, successfully handling all requests under any conditions	The additional feature's backend API implementation is provided and fully functional, handling requests under typical conditions	The additional feature's backend API implementation is provided and somewhat functional, handling requests under typical conditions	The additional feature's backend API implementation is weak, leading to common errors or unexpected results	No backend implementation of additional feature provided.
Error Handling (10%)	Explanation (5%)	The importance of both error-handling mechanisms are clear and well elaborated, and supported with strong examples	The importance of both error-handling mechanisms are explained with a weak example	The importance of both error-handling mechanisms are clearly explained	The importance of both error-handling mechanisms are briefly elaborated but clear	The importance of both error-handling mechanisms are not relevant or unclear Only one error handling mechanism is mentioned
	Implementation (5%)	The API recovers well from all types of errors or abnormal inputs, giving clear, actionable	The API recovers well from most errors or abnormal inputs, but not all messages are clear or helpful	The API can handle abnormal inputs , but does not provide meaningful error messages when error occurs	The API can handle simple errors but does not provide meaningful error messages	One or zero error handling mechanisms implemented

		error messages for various scenarios				
		Implementation for each error-handling mechanism is thoroughly explained and shows strong understanding of the code	Implementation for each error-handling mechanism is well explained and shows strong understanding of the code	Implementation for each error-handling mechanism is briefly explained but shows strong understanding of the code	Implementation for each error-handling mechanism is briefly explained and shows some understanding of the code	Explanation for code implementation is poor and shows clear lack of understanding
Demonstration & Code Interview (20%)	Q&A (20%)	Answers all technical/code-related questions confidently; gives a coherent explanation of logic and rationale behind technical solution. Present within the given time-limit.	Answers most technical/code-related questions confidently; gives reasonable explanation of logic and rationale behind technical solution. Present within the given time-limit.	Struggles with some technical/code-related questions; explanations of technical solution lacks depth or clarity.	Unable to respond to most technical/code-related questions; shows limited understanding of own technical solution.	Student was absent for the project demonstration without a valid reason. Clear evidence of plagiarism detected (disciplinable offence)

		Present within the given time-limit.				